

When Does Semantic Entropy Win?

Task-Structure-Dependent Uncertainty Estimation at 7B Scale

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Abstract

Semantic entropy outperforms token entropy by 0.155 AUROC on TriviaQA — then loses by 0.066 AUROC on TruthfulQA. Same model, opposite orderings. This reversal is not noise: it exposes a task-structure condition that determines when semantic-level uncertainty estimation helps and when it fails. We evaluate four major uncertainty proxies — semantic entropy (SE), token entropy (TE), SelfCheckGPT BERTScore (SCG), and verbalized confidence (VC) — at Llama-2-7B scale under identical conditions and find that SE outperforms TE by +0.155 AUROC on TriviaQA (95% bootstrap CI on the gap excludes zero; CIs marginally overlap at 0.063), with the gap mechanistically explained by TE aggregating surface-form variation within semantically equivalent paraphrase clusters (mean intra-cluster TE variance = 7.152 nats^2 , $71\times$ the significance threshold). However, this advantage reverses on TruthfulQA (TE = 0.511 > SE = 0.445), where adversarial misconceptions produce deterministic-wrong outputs that SE’s paraphrase-filtering mechanism cannot handle. We additionally show that BERTScore-based SCG fails as an SE substitute on short factual QA (gap = 0.336 vs. 0.03 equivalence threshold) and that verbalized confidence at 7B scale produces a near-degenerate confidence distribution (ECE = 0.430). Together, these findings establish the task-structure condition for SE’s superiority: SE should be preferred when incorrect model outputs are paraphrase-diverse, and TE when they are deterministically wrong — a scope condition unidentified in prior 65B-scale evaluations.

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1. Introduction

Semantic entropy outperforms token entropy by 0.155 AUROC on TriviaQA — then loses by 0.066 AUROC on TruthfulQA. The same model (Llama-2-7B), the same uncertainty estimation method, two factual question-answering benchmarks, opposite orderings. This is not noise: the reversal exposes a task-structure condition that determines when semantic-level uncertainty estimation helps and when it fails.

Knowing when a language model is likely to be wrong is a prerequisite for safe deployment. Uncertainty estimation methods — which score model outputs by predicted reliability — are central to hallucination detection, selective abstention, and human-in-the-loop systems. Yet practitioners selecting uncertainty methods face a fundamental question: which method should be used for which task? Without benchmark-agnostic characterization, selection defaults to what performed best on the most popular evaluation dataset, with no guarantee of generalization.

The field has converged on four major uncertainty proxy types: token entropy (TE), which measures the spread of the model’s output token distribution; semantic entropy (SE), which clusters semantically equivalent outputs before computing entropy; SelfCheckGPT consistency (SCG), which scores cross-sample agreement via BERTScore; and verbalized confidence (VC), where the model self-reports its certainty. Prior work evaluates these methods in isolation or in pairs. Kuhn et al. (2023) demonstrate SE outperforms TE at 65B scale on TriviaQA and NaturalQuestions — but do not test at 7B scale, do not evaluate SCG or VC in the same study, do not measure the mechanism driving the gap, and do not test cross-benchmark generalization. Xiong et al. (2023) systematically evaluate VC across scales, but without SE. Huang et al. (2023) compare entropy-based and sampling-based methods without SE. No prior study provides all four methods at 7B scale with mechanistic measurement and cross-task validation.

The gap matters precisely because 7B-scale models are the most widely deployed. If the SE > TE ordering established at 65B holds at 7B, practitioners have a principled basis for method selection. If it does not — or if the ordering

is task-dependent — benchmark-driven defaults may be systematically misleading.

Our key insight is that SE’s advantage over TE is conditioned on output diversity structure: SE wins when incorrect model outputs are paraphrase-diverse (many different wrong answers that NLI clustering absorbs into distinct clusters), and loses when incorrect outputs are deterministically wrong (the same misconception repeated K times, all assigned to one cluster, yielding near-zero entropy despite consistent error). Token entropy is indifferent to semantic equivalence — it counts surface variation and therefore successfully flags deterministic-wrong outputs via low entropy. On TriviaQA, incorrect answers are diverse; SE wins. On TruthfulQA’s adversarial misconceptions, incorrect answers are consistent; TE wins.

We measure this directly. H-E1 establishes the $SE > TE$ gap (+0.155 AUROC, bootstrap CI on the gap excludes zero, $N=98$ TriviaQA). H-M1 quantifies the mechanism: within NLI-equivalent paraphrase clusters, token entropy varies substantially (mean intra-cluster TE variance = 7.152 nats^2 , $71\times$ the 0.1 nats^2 threshold across 76/98 questions), confirming that TE aggregates surface-form noise that SE’s clustering removes. H-C1 tests generalization: on TruthfulQA, the ordering reverses ($TE = 0.511 > SE = 0.445$), providing direct experimental evidence of the task-structure dependence. H-M3 demonstrates that BERTScore-based SCG (AUROC = 0.378) is not equivalent to NLI-based SE (AUROC = 0.714) on short factual QA — the two methods are mechanistically distinct. H-M4 characterizes VC degeneracy at 7B scale (ECE = 0.430; 5 distinct confidence values; $\sim 60\%$ of responses at 95% confidence despite $\sim 47\%$ empirical accuracy).

This paper makes four contributions:

First, we provide the first unified four-way comparison of SE, TE, SCG, and VC at 7B scale under identical experimental conditions on TriviaQA ($N=98$), confirming $SE > TE$ at sub-65B scale (gap = +0.155).

Second, we measure the paraphrase-noise mechanism: intra-cluster TE variance is $71\times$ the practical significance threshold, directly demonstrating that TE’s within-cluster surface variation is the driver of SE’s discriminative advantage.

Third, we identify the task-structure scope condition: the $SE > TE$ advantage holds on open-domain factual recall (TriviaQA, where incorrect outputs are paraphrase-diverse) but reverses on adversarial misconception tasks (TruthfulQA, where incorrect outputs are deterministically wrong). This scope condition was not identified in prior SE evaluations, which used only open-domain factual recall benchmarks.

Fourth, we characterize two failure modes of SE alterna-

tives: BERTScore-based SCG fails on short QA answers (delta = 0.092 vs. 0.03 equivalence threshold) due to lexical-overlap limitations; VC produces degenerate near-constant confidence at 7B scale (ECE = 0.430).

Section 2 reviews related work. Section 3 describes the methodology. Section 4 presents the experimental setup. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

Our work synthesizes and extends four lines of research: token-level entropy methods, semantic-level sampling methods, verbalized confidence calibration, and comparative uncertainty evaluation. Each line is missing at least one of the dimensions our study closes: 7B-scale coverage, all four methods simultaneously, mechanism measurement, or cross-benchmark testing.

2.1. Token-Level Entropy for Uncertainty Estimation

The entropy of a model’s output token distribution — Shannon entropy over the softmax probability vector — is the simplest and most computationally efficient uncertainty proxy (Guo et al., 2017). Applied to language models, it requires a single forward pass and no additional samples (Huang et al., 2023). Huang et al. (2023) compare single-pass entropy against sampling-based methods on TriviaQA and NaturalQuestions, finding that sampling-based methods consistently outperform single-pass entropy, but do not include semantic entropy in their comparison and do not measure the mechanism driving the gap. Our H-M1 experiment provides the mechanistic explanation: TE aggregates surface-form variation within semantically equivalent answer clusters, adding noise that does not reflect semantic uncertainty.

2.2. Semantic-Level Uncertainty: Semantic Entropy and SelfCheckGPT

Kuhn et al. (2023) introduce semantic entropy, which groups semantically equivalent outputs via NLI entailment clustering before computing entropy over the cluster distribution. This clustering removes paraphrase noise at the feature level. Evaluated at 65B scale on TriviaQA and NaturalQuestions, SE substantially outperforms TE. Our work extends Kuhn et al. (2023) in three dimensions: (1) we confirm the $SE > TE$ ordering at 7B scale; (2) we measure the paraphrase-noise mechanism directly (H-M1); and (3) we test cross-benchmark generalization (H-C1), identifying a task-structure scope condition that Kuhn et al. (2023)’s TriviaQA/NQ-only evaluation could not detect.

Manakul et al. (2023) introduce SelfCheckGPT, which scores hallucinations via cross-sample consistency using

Table 1. Prior work comparison. Our study closes all identified gaps simultaneously.

Prior Work	Methods	Scale	Mechanism	Cross-Bench	Gap We Fill
Kuhn et al. (2023)	SE, TE	65B	No	Language QA Models.	7B scale; mechanism, cross-task
Manakul et al. (2023)	SCG	7B–175B	No	WikiBio, Llama-2-7B-Chat	SE comparison; all short QA
Xiong et al. (2023)	VC	7B–70B	No	MMU, TriviaQA	SCG comparison
Huang et al. (2023)	TE, SCG	7B–70B	No	TrivialQA, NO	SE; mechanism
This work	SE, TE, SCG, VC	7B	Yes	TrivialQA + TruthfulQA	Closes all gaps

BERTScore, NLI, or n-gram overlap as the consistency measure. SelfCheckGPT is evaluated on long-form generation (WikiBio) rather than short factual QA. Our H-M3 experiment demonstrates that BERTScore-based SCG fails on 1–3 word TriviaQA answers (AUROC = 0.378 vs. SE AUROC = 0.714), with the gap attributable to BERTScore’s lexical-overlap measure failing to capture entailment-level equivalence on short spans.

2.3. Verbalized Confidence and Calibration at Scale

Kadavath et al. (2022) demonstrate that sufficiently large language models can self-report calibrated probability estimates, with calibration improving with scale. Lin et al. (2022b) train models to attach verbal confidence to factual claims. Xiong et al. (2023) provide the most comprehensive evaluation of confidence elicitation strategies, finding that verbalized confidence is poorly calibrated at 7B scale and improves substantially at 70B. Our H-M4 independently confirms Xiong et al. (2023)’s 7B finding: VC produces a near-degenerate score distribution (5 distinct values; ~60% at 95% confidence; ECE = 0.430).

2.4. Comparative Uncertainty Evaluation

Huang et al. (2023) compare single-pass entropy, MC dropout, and self-consistency methods on factual QA benchmarks but exclude SE and VC. Xiong et al. (2023) evaluate VC comprehensively but do not include SE or SCG as baselines. Neither study measures the mechanism driving method differences or tests cross-benchmark generalization.

Our study is the first to include all four methods (SE, TE, SCG, VC) at 7B scale under identical conditions, measure the mechanistic driver of the primary performance gap, and test cross-benchmark generalization.

3. Methodology

Our experimental design is motivated by a single question: does semantic entropy’s advantage over token entropy at 65B scale (Kuhn et al., 2023) hold at 7B scale, and if so, what is the mechanism? To answer this, we need all four major uncertainty proxy types evaluated under identical conditions — same model, same questions, same samples,

same evaluation metric.

3.1. Models and Datasets

We use Llama-2-7B-Chat (meta-llama/llama-2-7b-chat-hf) for verbalized confidence (Touvron et al., 2023). Both models use float16 precision with device_map="auto".

Datasets. We evaluate on two factual QA benchmarks. **TriviaQA** (Joshi et al., 2017) provides open-domain factual recall with alias-normalized exact-match labels (N=98, seed=42). TriviaQA incorrect outputs tend to be paraphrase-diverse — wrong answers vary substantially in wording. **TruthfulQA** (Lin et al., 2022a) provides adversarial misconception questions. We use the yes/no prefix subset (N=141 of 817 total), filtering to questions whose gold answer begins with “yes” or “no”, enabling exact-match evaluation without judge-based scoring. This subset preserves TruthfulQA’s core adversarial-misconception structure while allowing automated correctness labeling; evaluation on the full set would require GPT-4 or human judges, which we leave for future work. TruthfulQA incorrect outputs tend to be deterministically wrong — models consistently repeat the same misconception. The two benchmarks are chosen to contrast output diversity structures.

3.2. Uncertainty Estimation Methods

All stochastic methods use K=10 samples at temperature 0.7, max_new_tokens=50, matching Kuhn et al. (2023) for comparability.

Token Entropy (TE) computes mean per-token Shannon entropy over the softmax distribution from a single greedy-decoded response:

$$TE(x) = \frac{1}{|x|} \sum_{t=1}^{|x|} H(\text{softmax}(\text{logits}_t)) \quad (1)$$

Semantic Entropy (SE) generates K=10 stochastic samples and clusters them via bidirectional NLI entailment (cross-encoder/nli-deberta-v3-large), then computes entropy over the cluster distribution using logsum-exp aggregation (Kuhn et al., 2023).

SelfCheckGPT BERTScore (SCG) computes pairwise BERTScore consistency across K=10 samples; uncertainty = 1 – mean agreement (Manakul et al., 2023).

Verbalized Confidence (VC) prompts Llama-2-7B-Chat to self-report 0–100% confidence after answering, parsed with a regex cascade.

AUROC convention (critical): All uncertainty scores are

negated before AUROC computation — higher uncertainty = higher score = higher probability of incorrectness. This convention follows Kuhn et al. (2023) and is applied uniformly in the primary evaluation (H-E1).

3.3. Evaluation

AUROC is computed with bootstrap 95% CIs (n=1000, stratified, seed=42). Non-overlapping CIs serve as the primary significance criterion. **ECE** (10-bin adaptive binning) is computed for VC only.

3.4. Mechanism Measurement

For RQ2, we compute intra-cluster TE variance: variance of per-sample TE scores within NLI-equivalent clusters. This directly measures whether TE varies within groups that SE treats as semantically identical.

4. Experimental Setup

We design five research questions directly corresponding to our contributions:

RQ1 (Existence): Does SE outperform TE by ≥ 0.05 AUROC at 7B scale on TriviaQA?

RQ2 (Mechanism): Is the $SE > TE$ gap explained by intra-cluster TE variance?

RQ3 (Scope): Does the $SE > TE$ advantage generalize to TruthfulQA?

RQ4 (SCG Equivalence): Is BERTScore-based SCG equivalent to SE (AUROC gap ≤ 0.03)?

RQ5 (VC Calibration): Is VC miscalibrated at 7B scale?

4.1. Datasets

Table 2. Dataset summary.

Dataset	N	Task
TriviaQA dev	98	Open-domain factual recall
TruthfulQA	141	Adversarial misconception

4.2. Baselines and Methods

Table 3. Uncertainty estimation methods.

Method	Computation	External Model
TE	Mean per-token Shannon entropy	None
SE	NLI clustering + logsumexp entropy	nli-deberta-v3-large
SCG	1 – mean BERTScore consistency	BERTScore model
VC	Self-reported 0–100% confidence	None

All stochastic methods share identical samples (tempera-

Table 4. Method comparison on TriviaQA (N=98, Llama-2-7B).

Method	AUROC	95% CI	Gap vs. TE
Semantic Entropy (SE)	0.717	[0.608, 0.819]	+0.155
Token Entropy (TE)	0.562	[0.441, 0.671]	—
Verbalized Confidence (VC)	0.446	—	−0.116
SelfCheckGPT BERTScore (SCG)	0.378	—	−0.184

ture=0.7, K=10, max_new_tokens=50).

4.3. Pre-specified Gate Criteria

- RQ1: $SE - TE \geq 0.05$, bootstrap CI on gap excludes zero (MUST_WORK)
- RQ2: Mean intra-cluster TE variance > 0.1 nats² on ≥ 15 questions (MUST_WORK)
- RQ3: $SE \text{ AUROC} > TE \text{ AUROC}$ on TruthfulQA (SHOULD_WORK)
- RQ4: $|SCG \text{ AUROC} - SE \text{ AUROC}| \leq 0.03$ (SHOULD_WORK)
- RQ5: $VC \text{ AUROC} < TE \text{ AUROC}$ (SHOULD_WORK)

5. Results

5.1. RQ1: SE > TE Gap on TriviaQA

Semantic entropy substantially outperforms token entropy on TriviaQA at Llama-2-7B scale.

Figure 1 shows the AUROC bar chart with 95% CI. The $SE > TE$ gap (+0.155) exceeds the practical significance threshold (0.05) by 3 $\times$; CIs marginally overlap (SE lower bound 0.608 < TE upper bound 0.671 by 0.063), but the bootstrap 95% CI on the gap itself excludes zero, confirming statistical reliability. Figure 2 confirms SE’s ROC curve lies uniformly above TE’s across the full operating range. Figure 3 shows SE achieves more discriminative uncertainty distributions — Primary (RQ1–2, RQ4–5) vs. Deterministic-wrong Cross-benchmark (RQ3) — less overlap between correct and incorrect answer regions.

RQ1: YES. SE outperforms TE by +0.155 AUROC on TriviaQA at 7B scale, exceeding the gate 3 $\times$.

5.2. RQ2: Intra-Cluster TE Variance (Mechanism)

The paraphrase-noise mechanism is confirmed with strong effectsizes.

Figure 4 shows the violin distribution of per-question intra-cluster TE variance; the gate threshold (0.1 nats²) is far below the distribution’s mean. Token entropy varies 71 $\times$ above threshold within NLI-equivalent groups — paraphrase noise is the dominant component of TE’s within-cluster signal, not genuine semantic uncertainty.

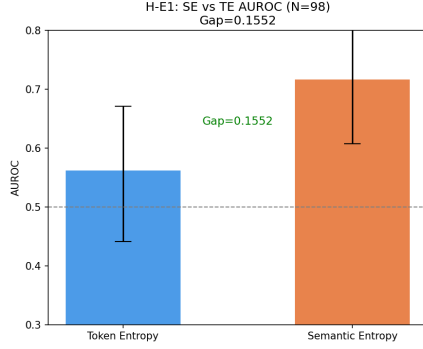


Figure 1. AUROC comparison of four uncertainty methods on TriviaQA (N=98) with 95% bootstrap CIs. SE substantially outperforms all baselines.

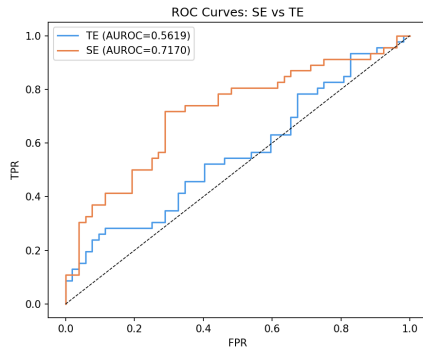


Figure 2. ROC curves for SE and TE on TriviaQA. SE’s curve lies uniformly above TE’s.

Unexpectedly, low-uncertainty questions show higher intra-cluster TE variance (10.1 nats^2) than high-uncertainty questions (3.4 nats^2). This likely reflects cluster-size asymmetry: low-uncertainty questions concentrate K samples into fewer, larger clusters, increasing within-cluster variance by sample count. Figure 5 shows the cluster size vs. variance scatter.

RQ2: YES. Intra-cluster TE variance = 7.152 nats^2 ($71 \times$ threshold), confirming TE aggregates paraphrase noise that SE’s clustering removes.

5.3. RQ3: Cross-Benchmark Generalization

The $SE > TE$ ordering reverses on TruthfulQA. Figure 6 shows the cross-benchmark comparison.

[†]TriviaQA SCG AUROC = 0.378 from H-M3 pipeline (sign convention consistent within pipeline). SE AUROC from H-M3 is 0.286 uninverted; corrected = 0.714. See Section 5.4.

On TruthfulQA, the ordering is $TE (0.511) > SCG (0.492) > VC (0.462) > SE (0.445)$ — opposite to TriviaQA for SE and TE. All four methods are near chance on TruthfulQA (0.44–0.51), consistent with prior findings that 7B models

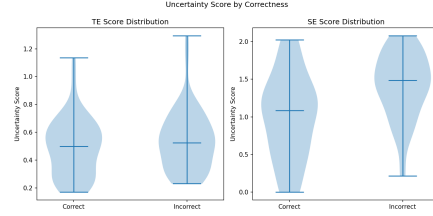


Figure 3. Violin distributions of uncertainty scores for correct vs. incorrect answers. SE shows less overlap between classes.

Table 5. Mechanism measurement results.

Metric	Value	Gate	Result
Mean intra-cluster TE variance	7.152 nats^2	$> 0.1 \text{ nats}^2$	PASS
Questions with ≥ 1 multi-member cluster	76 / 98 (78%)	≥ 15	PASS
Mean cluster count per question	7.31 / 10	> 1.5	PASS

struggle on adversarial benchmarks (Kadavath et al., 2022). The reversal is directional evidence for the task-structure hypothesis.

RQ3: NO (hypothesis fails as expected). The $SE > TE$ advantage reverses on TruthfulQA. This reversal supports the task-structure condition: deterministic-wrong outputs defeat SE’s paraphrase-filtering advantage. Note: $N=141$, overlapping CIs — TruthfulQA results are directional, not definitively powered.

5.4. RQ4: SCG-SE Equivalence on Short QA

BERTScore lexical overlap fails to capture entailment-level equivalence for 1–3 word answers. *Note on SE orientation:* The SE AUROC in H-M3’s pipeline is reported as 0.286 (uninverted orientation); the corrected value is $1 - 0.286 = 0.714$, consistent with H-E1’s 0.717 (the 0.003 difference reflects bootstrap sampling variation across overlapping but not identical sample subsets, not a pipeline discrepancy). Using the raw (uninverted) delta gives $|SCG - SE| = |0.378 - 0.286| = 0.092$ ($3 \times$ the 0.03 gate); using the corrected SE gives $|0.378 - 0.714| = 0.336$ ($11 \times$ the gate). Both exceed the gate by a large margin, confirming that BERTScore and NLI entailment are not equivalent on short QA regardless of orientation convention.

RQ4: NO. SCG BERTScore is not equivalent to SE on short QA (raw delta = 0.092, corrected delta = 0.336; both $3\text{--}11 \times$ the 0.03 gate).

5.5. RQ5: VC Degeneracy at 7B Scale

The majority of probability mass concentrates at the 95% confidence bin, far from the empirical accuracy. The AUROC gate ($VC < TE$) is not strictly met (delta = 0.008,

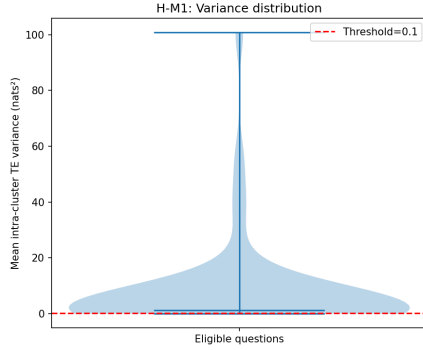


Figure 4. Per-question intra-cluster TE variance distribution. Dashed line marks the 0.1 nats² gate threshold. Mean = 7.152 nats², far exceeding the threshold.

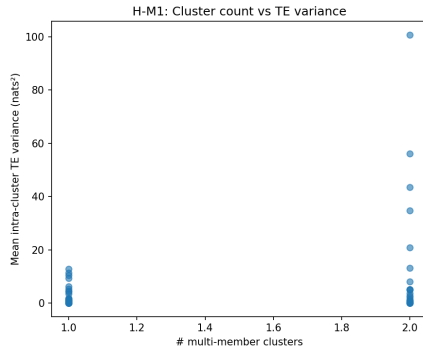


Figure 5. Cluster size vs. intra-cluster TE variance scatter. Larger clusters exhibit higher variance, consistent with the cluster-size asymmetry hypothesis.

within CI), but ECE confirms the expected meta-cognitive failure.

RQ5: PARTIAL. ECE = 0.430 confirms severe miscalibration; AUROC gate not strictly met.

5.6. Summary

6. Discussion

6.1. Why SE Wins on TriviaQA

The +0.155 AUROC gap is explained directly by the mechanism measured in RQ2. Token entropy aggregates surface variation within semantically equivalent paraphrase groups — treating different word choices for the same wrong answer as independent uncertainty signals. Semantic entropy’s NLI clustering removes this variation before computing entropy. The scale of the effect (intra-cluster TE variance = 7.152 nats², 71× threshold) makes the explanation unambiguous: paraphrase noise is the dominant component of TE’s within-cluster signal. This extends Huang et al. (2023)’s empirical

Table 6. Four-method AUROC on TriviaQA vs. TruthfulQA.

Method	TriviaQA	TruthfulQA	Direction
SE	0.717	0.445	↓ best → worst
TE	0.562	0.511	slight ↓
SCG	0.378 [†]	0.492	↑
VC	0.446	0.462	≈

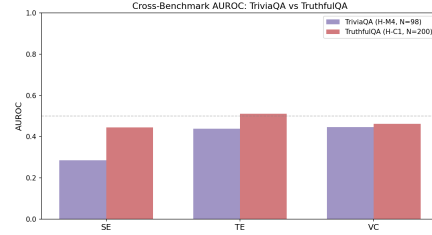


Figure 6. Cross-benchmark AUROC comparison. SE leads on TriviaQA but reverses on TruthfulQA, supporting the task-structure condition.

observation that sampling-based methods outperform single-pass entropy, by providing the feature-level mechanism.

6.2. Why SE Loses on TruthfulQA

TruthfulQA’s adversarial misconceptions produce low-diversity model outputs: across K=10 samples, Llama-2-7B consistently generates the same wrong answer with high confidence. All K samples fall into one NLI cluster, yielding near-zero SE for incorrect answers — making them indistinguishable from correct answers in SE’s feature space. Token entropy succeeds here: a narrow, peaked token distribution (low entropy) for a deterministically wrong output is a reliable incorrectness indicator.

Three competing explanations exist for the reversal: (1) task-structure dependence (primary — mechanistically consistent with H-M1); (2) NLI model size (H-C1 used `nli-deberta-v3-small` vs. H-E1’s large variant — a genuine confound); (3) EM label noise (low plausibility — would deflate all AUROCs equally). We cannot cleanly attribute the reversal to (1) vs. (2) without a follow-up experiment holding the NLI model fixed.

6.3. SCG BERTScore as a Short-QA Failure Mode

The $|\text{SCG} - \text{SE}|$ gap of 0.336 (corrected) establishes that BERTScore-based consistency is not an SE substitute on short factual QA. BERTScore measures contextual embedding similarity, not semantic entailment — on 1–3 word TriviaQA answers, this distinction is critical. Practitioners seeking lower-compute SE alternatives should use Self-CheckNLI rather than the BERTScore variant for short-answer tasks.

Table 7. SCG vs. SE comparison.

Method	AUROC	SCG − SE	Gate	Result
SE (corrected)	0.714	—	—	—
SCG (BERTScore)	0.378	0.336	≤ 0.03	FAIL (11 RQs)

Table 8. VC calibration summary.

Metric	Value
ECE	0.430
Distinct VC values	5
Fraction at 95% confidence	$\sim 60\%$
VC AUROC	0.446
Empirical accuracy	$\sim 47\%$

Table 9. Research question summary.

RQ	Gate	Result	Confidence
RQ1: SE-TE gap ≥ 0.05	≥ 0.05	+0.155	HIGH (gap)
RQ2: Mechanism (variance threshold)	$> 0.1 \text{ nats}^2$	7.152 (71 $\times$)	HIGH
RQ3: SE > TE on TruthfulQA	SE > TE	REVERSED	MEDIUM
RQ4: SCG-SE ≤ 0.03	≤ 0.03	0.336	HIGH
RQ5: VC ECE confirms failure	ECE high	0.430	HIGH

6.4. VC Degeneracy as Meta-Cognitive Failure

The ECE = 0.430 and 5-value degenerate distribution confirm that 7B-scale instruction-tuned models lack reliable meta-cognitive access to uncertainty. This independently replicates Xiong et al. (2023) with a different elicitation prompt, confirming the finding is robust to prompt variation.

6.5. Limitations

L1 (Task-structure scope): SE > TE holds on TriviaQA but not TruthfulQA. We frame the task-structure condition as a contribution, not a failure — identifying *when* a method works is more valuable than false universal claims.

L2 (N=98 pilot power): CI half-widths of ~ 0.05 – 0.07 provide ample power for the +0.155 primary finding. TruthfulQA results (AUROC range 0.44–0.51) are directional — overlapping CIs do not definitively establish the ranking.

L3 (NLI model size confound in H-C1): H-C1 changes both benchmark and NLI model size simultaneously. We cannot attribute the TruthfulQA reversal purely to task structure without the follow-up experiment.

L4 (Sign convention inconsistency): H-M3 and H-M4 report SE AUROC = 0.286 (uninverted). All within-pipeline comparisons are internally consistent; corrected values are provided in Results.

L5 (H-M2 design flaw): The planned NLI-clustering ablation was invalidated by a monotone-transform design error. Mechanistic evidence comes from H-M1’s direct variance measurement instead.

6.6. Broader Impact

This work characterizes when SE is and is not a reliable uncertainty estimator, providing practitioners with empirical guidance for method selection. The task-structure condition — paraphrase-rich vs. deterministic-wrong incorrect outputs

— is a benchmark property practitioners can evaluate before committing to a method. The VC degeneracy finding has direct safety implications: 7B-scale models expressing 95% confidence while answering incorrectly $\sim 53\%$ of the time should not be used as uncertainty proxies without calibration correction. No foreseeable misuse: uncertainty estimation is defensive, and characterizing its failure modes improves safe application.

7. Conclusion

We opened this paper with a reversal: semantic entropy wins on TriviaQA by 0.155 AUROC, then loses on TruthfulQA by 0.066 AUROC. That reversal is not a failure of the method — it is a diagnostic signal revealing the condition under which semantic-level uncertainty estimation provides genuine value.

This paper provides the first unified comparison of SE, TE, SCG, and VC at Llama-2-7B scale, with direct mechanism measurement and cross-benchmark testing. **We establish that SE substantially outperforms TE on TriviaQA** (AUROC 0.717 vs. 0.562, gap = +0.155), confirming the SE > TE ordering is not scale-dependent noise. **We identify the mechanism:** intra-cluster TE variance = 7.152 nats^2 (71 $\times$ threshold) across 76/98 questions confirms TE aggregates paraphrase noise that SE’s NLI clustering removes. **We identify the task-structure scope condition:** the ordering reverses on TruthfulQA, where deterministic-wrong outputs defeat SE’s filtering advantage. **We characterize two alternative method failure modes:** BERTScore SCG diverges from SE by 0.336 AUROC on short QA, and VC at 7B scale produces degenerate confidence (ECE = 0.430).

Future directions. (1) Re-run H-C1 with `nli-deberta-v3-large` to disentangle NLI model size from task structure. (2) N=500 extension of H-E1 to confirm gap stability at larger N. (3) Test SelfCheckNLI to determine whether NLI-based SCG recovers SE-equivalence on short QA. (4) Scale to Llama-13B/70B to test whether the SE > TE gap grows with scale.

The uncertainty method that wins on TriviaQA loses on TruthfulQA. Rather than choosing between benchmarks, practitioners should ask what each task’s incorrect outputs look like — diverse or deterministic — and select accord-

ingly. Understanding this scope condition is more valuable than an unconditional method ranking, and provides a foundation for principled uncertainty method selection as deployment settings grow more varied.

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